

# Cyclone Detection and Intensity Estimation using Satellite Images

## B. Tech. Project Report

*Submitted by*

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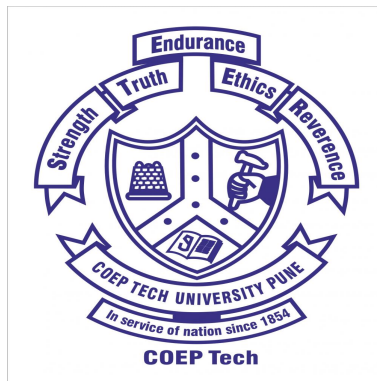
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# Abstract

Tropical cyclones are among the most devastating natural disasters, causing significant loss of life and property globally. Accurate detection, tracking, and intensity estimation are crucial for disaster preparedness and response. Traditional methods like the Dvorak technique depend on manual satellite image interpretation, which is time-consuming and subjective. Recent advances in remote sensing and deep learning (DL) have led to automated, data-driven cyclone analysis methods. This research presents a deep learning framework that integrates multi-modal satellite data fusion with lightweight architectures for improved cyclone detection and intensity estimation. Leveraging convolutional neural networks (CNNs) and hybrid models, the system enables real-time tracking and prediction, addressing challenges like data heterogeneity, model generalization, and processing speed. The approach achieves lower root mean squared error (RMSE) in intensity estimation and improves cyclone detection. Techniques like super-resolution and optical flow further enhance feature extraction from low-resolution satellite images. By blending insights from earlier studies with modern models, the research delivers a scalable and efficient solution, supporting meteorologists and emergency responders with timely and accurate information. This work highlights the transformative role of deep learning in advancing cyclone tracking and disaster management.

**Keywords:** Deep learning, wind speed estimation, Convolutional neural network, Estimation, Cyclone, Tropical Cyclone Detection, Disaster management

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# Chapter 1

## Introduction

Tropical cyclones are one of the most disastrous natural hazards and result in unprecedented loss of property and lives all over the world. Proper detection, tracking, and intensity measurement of cyclones are essential to prepare for and respond to a disaster. These tasks are commonly performed using methods like the Dvorak method [1] that are purely based on the manual interpretation of satellite images. These are both time-consuming, subjective, and limited in analysis.

Conventional approaches based on human interpretation of satellite data [2] are constrained by latency and subjectivity. Early studies pioneered the use of satellite data for cyclone genesis detection [3] and center localization [4]. Subsequent advancements introduced CNNs for intensity estimation [5] and hybrid models for multi-sensor fusion [6]. Recent innovations leverage cutting-edge architectures like Faster R-CNN and YOLOv8 for real-time tracking [7], while super-resolution techniques enhance feature extraction from low-resolution satellite data.

With the introduction of remote sensing technologies and deep learning, there has been a shift towards data-driven and automatic methods of cyclone analysis. The integration of artificial intelligence (AI), particularly deep learning (DL), has emerged as a transformative solution, enabling automated,

real-time, and high-accuracy frameworks for cyclone Detection . [6] [8] [9]

Still challenges persist in spite of advances. Data integration with heterogeneous data [10], model generalization over regions [11], and real-time processing requirements call for more innovation. This research fills these holes by introducing a new DL framework that integrates multi-modal satellite data fusion with light-weight architectures for effective cyclone detection and intensity estimation. By combining knowledge from pioneering articles [12] and cutting-edge models [13], our solution seeks to provide improved accuracy and scalability for cyclone tracking systems.

This paper introduces a diagnostic tropical cyclone intensity estimation system that leverages satellite imagery and machine learning algorithms to provide timely and accurate predictions of cyclone intensity levels. The core of the system lies in the implementation of a convolutional neural network (CNN) model trained on historical cyclone data to estimate intensity levels based on features extracted from satellite imagery preprocessing the imagery and feeding it into the CNN model, the system generates objective intensity estimations, enabling meteorologists, emergency responders, and policymakers to make informed decisions in anticipation of cyclone events. [14]

# Chapter 2

## Literature Review

Over the past decade, many researchers in the field of machine learning have experienced major leaps. The increased negative effects of tropical cyclones and increased awareness of the catastrophic impacts on coastal communities have stimulated the development of advanced prediction systems. The catastrophic consequences of tropical cyclones related to infrastructure, economy and human life have made accurate and early predictions extremely important. Many researchers recognize the need for better predictive tools and limited confidence in population awareness, and are faced with machine learning (ML) and deep-learning techniques to improve cyclone intensity and prediction of tragic effects.

Uma et al. has provided a complete deep learning-based method is provided to predict and estimate real-time and spontaneous tropical cyclones utilizing wind velocity. The study achieves a better Root Mean Squared Error (RMSE), demonstrating the effectiveness of Convolutional Neural Networks (CNNs) in cyclone intensity estimation. This study also shows that combining spatial feature extraction (CNN) enhances prediction accuracy. Leveraging data from MOSDAC site and incorporating with INSAT-3D satellite imagery, the approach achieves higher precision, predicting wind speeds at 1-knot (kt).

The previous study uses traditional methods like the Dvorak Technique, Kalman Filters, and Machine Learning (ML) models such as Support Vector Machines (SVM), Decision Trees (DT), and Random Forest (RF) for cyclone classification, achieving accuracy ranging from 86% to 95% [15].

Patil et al. constructed a Deep Learning object detection system was constructed to locate a cyclone's eye, achieving an accuracy of over 87%. The CNN model attained over 95% accuracy, while other models achieved:

- 94% for SVM,
- 93% for RF,
- 93% for VGG16,
- 91% for XGBoost,
- 88% for Artificial Neural Networks (ANN), and
- 86% for the K-Nearest Neighbors (KNN) model.

Additionally, 98.42% and 98.23% accuracy were achieved using K-fold and Stratified K-fold cross-validation, respectively, for the CNN model. The SVM model achieved 99.66% and 98.84% accuracy. A mean Average Precision (mAP) of 87.86% was achieved for an average Intersection over Union (IoU) of 65.37% using YOLOv4 [16].

Liu et al. show that CNN+LSTM with infrared satellite imagery from the GOES-16 satellite, preprocessed using histogram equalization, resulted in 94% accuracy in detecting cyclones 12 hours earlier than traditional methods. Limitations include high computational cost for real-time processing. The study also used Vision Transformer (ViT) for multispectral images and MODIS data, augmented with synthetic cyclones, resulting in improved in-

tensity classification using self-attention, with an F1-score of 0.89 for Category 3+ cyclones [17].

Zhang et al. employs pre-trained Convolutional Neural Networks (CNNs) for tropical cyclone (TC) intensity estimation. Specifically, four CNN architectures were evaluated: VGGNet, GoogLeNet, ResNet, and Xception. An optimized CNN model, specifically the Visual Geometry Group Network (VGGNet), was ultimately obtained for TC intensity estimation. When applied to the test data, the model achieves a relatively low mean absolute error (MAE) of  $4.05 \text{ m s}^{-1}$ . The proposed model outperforms traditional TC intensity estimation techniques, such as the Dvorak technique and other machine learning models (e.g., PHURIE-SVR, Deep-PHURIE, and ADT9.0). By integrating temporal sequential images and auxiliary physical information, the MAE was reduced by 8% compared to models without this integration. The study used geostationary satellite infrared (IR) imagery from the Himawari series (GOES-9, MTSAT-1R, MTSAT-2, and Himawari-8) spanning 2004–2022 over the western North Pacific Ocean [18].

Keerthivasan et al. introduce a hybrid CNN-RNN model that leverages both spatial and temporal information for cyclone intensity estimation, addressing the limitations of traditional CNN-based approaches. The model achieves an accuracy of 96.7% and an RMSE of 3.58, indicating high precision in cyclone intensity estimation [19].

Kavitha et al. uses satellite imagery from Infrared (IR) and Water Vapor (WV) channels to detect and classify cyclones. The dataset includes 500,000 non-cyclone images and 50,000 cyclone images, which are preprocessed and labeled for training the model. The model achieves an accuracy of 95% in cyclone detection and eye localization [20].

Talukder et al. incorporates multiple data sources, including QuikSCAT

(wind data) and TRMM (precipitation data), to detect and track cyclones globally. QuikSCAT is used for accurate cyclone detection, while TRMM is used for fine temporal resolution tracking. The ensemble classifier achieves a true positive rate (TPR) of 91.67% and a true negative rate (TNR) of 76.07% in cyclone detection. The system successfully detects all 26 tropical cyclones reported by the National Hurricane Center (NHC) in 2005, with additional detections of subtropical storms and tropical depressions [21].

This study proposed a deep learning model for classifying and predicting cyclones using satellite images that combines CNNs and optical flow techniques. The study demonstrates how augmentation and interpolation practices increase model precision by ensuring higher temporal resolution for the training datasets. The model achieved 97% accuracy in classifying cyclones and more than 84% confidence in detecting cyclones.

Shakya et al. result suggest that RetinaNet has an advantage over LSTM for low-frequency datasets, while YOLO excelled for cyclone detection and R-CNN for predicting storm locations. This study also highlights the contribution of preprocessing methods, such as SFI optimization for error and optical flow estimation for interpolation, in achieving better results. Furthermore, linear regression was effectively used to forecast the cyclone path. Nevertheless, the study identified a difficulty in dealing with multiple cyclones in an image and recommends future work to include the effects of the Earth's curvature. These results show the effectiveness of deep learning techniques in improving cyclone detection and tracking, and further demonstrate reduced manual intervention while increasing accuracy and speed [22].

Xie et al. designed a cyclone detection scheme that integrated rainfall data from a precipitation radar and wind field vectors from ASCAT scatterometer using data fusion methodology. They argue that data from mul-

multiple remote sensing sources are more beneficial for cyclone detection than using only a single source. This model uses a Feature Pyramid Network (FPN) with Faster R-CNN to detect cyclones globally. The study’s ablative experiments showed that wind field data results in higher detection accuracy compared to precipitation data. This study does have some limitations such as the ‘cask effect’ where the spatial resolution of the dataset is limited to the lowest resolution data layer, making precise localization of the cyclone eye impossible. Regardless of this limitation, the study underscores the fact that deep learning-based systems for automated cyclone detection with data from different sources have great potential in improving remote sensing and data analysis of climate patterns [23].

One of the concerns that remains unresolved is how to best predict cyclone formation. Piñeros et al. used a statistical approach for detecting tropical cyclone genesis with the help of infrared (IR) satellite images and proposed a technique based on the quantification of axisymmetric cloud organization on clusters using cloud brightness-temperature gradient orientation analysis. The results indicated that stronger axisymmetry in cloud clusters was significantly associated with cyclone formation and could be identified about 0.6 hours prior to being classified as a tropical depression. While the proposed method achieved a true positive rate of 93%, it was accompanied by a 22% false alarm rate, which indicated that further refinement of the classification method is warranted. Additional classification criteria that help minimize false positives, as well as the development of the method into a real-time prediction system, are envisioned refinements [24].

Mohammed Nhj et al. presents a two-stage deep learning model capable of identifying tropical cyclones and locating their associated central cloud formations on satellite images. The model underscores the contribution of

deep learning towards cyclone detection and forecasting as it highlights the automation of detection. One of the main hurdles faced was the unavailability of a benchmark dataset, which impacted model training and testing. Regardless of this limitation, the model was effective in identifying cyclones. The researchers hypothesize that incorporating surface temperature and air pressure information in future work would enhance the reliability of cyclone tracking, forecasting, and center localization. This research affirms the essence of deep learning in cyclone detection and prediction, yet more exploration in its accuracy and general applicability is necessary [25].

Moustafa et al. offered an edge-enhanced super-resolution GAN (EESR-GAN) model which focuses on cyclone detection and boundary identification through deep learning. This fully automated detection network combines a generator with RRDB (Residual-in-Residual Dense Block) architecture and a Faster R-CNN discriminator to improve the quality of resolution from low-quality satellite pictures. The method successfully retrieves cyclone features disregarding the issues of resolution limitations. The research used the Community Atmospheric Model (CAM5.1) dataset and used four measures: precision, recall, intersection over union (IoU), and mean average precision (mAP). The results achieved 86.3% detection accuracy and an average precision of 88.63%, exceeding most averaged object detection models. This study showcases the capability of super-resolution methods in cyclone detection and proposes a faster and more efficient way to analyze extreme weather events for research purposes [26].

Harris et al. focused on retrieving information such as contention-based cache has presented Cyclone, a distributed anomaly detection launched to track cyclic interference, a common feature to all known contention-based cache attacks. Cyclone differs from traditional methods that focus on cache



partitioning and identifying contention anomalies by using cyclic interference for highly accurate detection of micro-architectural attacks with low false positive rates.

The proposed approach is efficient to scale vertically with multiple parallel domains within a single address space, making it very applicable to security for browsers and isolation at the core level. Results from experiments done within an ARM microarchitecture show close to ideal detection rates with  $260\text{--}1000\times$  lower rates of false positives as opposed to other contention-based detection methods, with negligible active measures cost ( $\sim 3.6\%$  performance degradation). This study opens up other possibilities for the protection of micro-architectural features, particularly in systems that need to enforce strong isolation across several security domains [27].

A convolutional neural network (CNN)-based methodology leveraging satellite pictures has been developed for evaluating the intensity of tropical cyclones (TC). The classical approach, like the Dvorak method, faces a lot of biases as well as differences among various basins. The CNN approach gets rid of these difficulties by using the infrared brightness temperature and the microwave rain rate for over 1097 TCs spread across the globe (2003–2014) with additional TC features like location, time, season, etc.

Chen et al. enhanced CNN model demonstrated 79% accuracy within a 10-knot error margin and achieved a root-mean-square error (RMSE) of 8.39 knots in independent testing on 94 global TCs (2017). Further improvements include incorporating surface temperature, air pressure data, and post-smoothing techniques to enhance forecasting precision. This study highlights the potential of CNNs in cyclone intensity estimation, offering a reliable and scalable alternative to traditional intensity estimation methods [28].

Uma et al. introduced a study of Tropical cyclones are among the most

financially damaging natural disasters, making accurate assessment crucial for minimizing losses and saving lives. Various methods use satellite data to measure cyclone intensity, but their effectiveness varies. This study applies deep learning to infrared satellite data, achieving an RMSE of 13.24 knots for cyclone strength prediction. Additionally, it introduces a first-of-its-kind visualization portal in a production system, offering valuable insights to users. [29]

Goyal et al. describe how India's coastal regions, especially Odisha, Andhra Pradesh, West Bengal, and Tamil Nadu, face frequent and intense cyclones, bringing strong winds, heavy rains, and floods. Climate change has only made things worse, putting millions of people at risk. While better warning systems have saved lives since the devastating 1999 Odisha super cyclone, the damage to homes, infrastructure, and livelihoods remains severe. To truly protect coastal communities, disaster risk reduction needs to be a priority in every part of development planning. [30]

Kaur et al. highlight that Hurricanes, or tropical cyclones, are among the most devastating natural disasters, causing massive destruction to communities, economies, and the environment. While social media now plays a crucial role in providing immediate relief, analyzing its vast data can be challenging. Satellite imagery helps track damage over large areas, but manual assessment is often inaccurate. To improve disaster response, machine learning and deep learning are being used to automate damage detection, making recovery efforts faster and more efficient. This paper explores how these technologies are transforming disaster management. [31]

Knapp et al. explain that geostationary satellites have provided continuous Earth observations since the 1970s, but their use in climate studies has been limited due to data volume, lack of a central archive, and incon-

sistent formats. The International Satellite Cloud Climatology Project (ISCCP) helped address these challenges by archiving geostationary data since 1983. NOAA’s National Climatic Data Center has further improved access by standardizing projections, recalibrating data, and extending records back to 1980. The Gridded Satellite (GridSat) dataset offers easy-to-use climate data in netCDF format, supporting research on tropical cyclones, global precipitation, and atmospheric patterns. [32]

Mawatwal and Das propose Predicting tropical cyclone intensity remains a challenge due to complex dynamics and limited data over the North Indian Ocean. This study presents a hybrid deep learning model using a Convolutional Neural Network (CNN) to analyze IR satellite imagery from 2000–2022. The model includes a binary classifier, multiclass classifier, YOLOv3-based cyclone detector, and a regression module. It achieved 98.4% binary classification accuracy, 63.83% multiclass accuracy, and an RMSE of 16.2 knots. The study highlights the impact of dataset size, time-series testing challenges, and the need for further data collection to improve cyclone prediction. [33]

Talukder et al. have developed an automated cyclone tracking system that utilizes images from multiple satellite sources. The system processes infrared images from the Geostationary Operational Environmental Satellite (GOES), precipitation data from five satellites, and ocean surface wind field images. It comprises three key components: data preprocessing, cyclone eye detection for each data source, and a filter-based tracker that integrates detection results. Experimental results demonstrate that the system is operationally viable and outperforms previous cyclone tracking models. [34]

Maskey et al. highlight that tropical cyclones cause immense damage worldwide, making accurate intensity prediction crucial for saving lives and property. While several techniques estimate cyclone wind speed using satel-

lite data, their effectiveness varies. This study introduces a deep learning model that predicts cyclone intensity from infrared satellite images with a 13.24-kn RMSE. Additionally, it features an innovative visualization portal that presents deep learning results alongside contextual data, making cyclone analysis more accessible for users. [35]

Maskey et al. explain that enterprises are increasingly integrating machine learning into their future strategies, and the Earth science domain is following suit. However, transitioning machine learning from research to real-world applications remains a challenge, as building a model differs significantly from deploying it for decision-making. This study explores the key phases of the machine learning lifecycle, focusing on the successful transition of models into production for practical use. [36]

# Chapter 3

## Research Gaps and Problem Statement

### 3.1 Research Gaps

Cyclone detection and classification have, historically, depended on methods like the Dvorak Technique and Kalman Filters, which have been complemented in recent years by machine learning methods like Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF). Even though the methods have been highly accurate, between 86% and 95%, they also have some drawbacks. Most of the conventional methods are based on manual analysis of satellite images, and this causes subjectivity and delay in decision-making. In addition, conventional machine learning models require a lot of feature engineering and may not capture the complex, nonlinear structures in raw satellite data effectively. Also, present-day models are not usually optimized for real-time cyclone detection and hence their usage in emergency forecasting and fast response scenarios is limited. There exists a definite void in the development of end-to-end automated, deep learning-based systems that have the capability to directly interpret satellite images to detect cyclones more accurately, timely, and with minimal human inter-

vention. Bridging this void can go a long way in augmenting early warning systems, disaster preparedness, and response planning.

## **3.2 Problem Statement and Objectives**

Traditional methods for cyclone detection often rely on manual analysis and heuristic approaches, which can be slow and inaccurate. Our study shows for more automated and efficient cyclone detection systems that take advantage of the vast amounts of satellite imagery and meteorological data available. Using deep learning approach like CNN for better cyclone intensity estimation and accurate prediction.

The objectives of this study are

- To make a survey of existing techniques that are proven sound enough to cyclone detection.
- Develop a Deep Learning-Based Cyclone Detection System
- Improve the prediction of the intensity Estimation as well as wind speed detection.
- Implement a real-time weather map to monitor and visualize wind speed patterns, enabling accurate detection of potential cyclone formation and enhancing early warning capabilities.

# Chapter 4

## Proposed Methodology

Figure 4.1 shows workflow of the proposed methodology, illustrating the steps involved in each part.

### 4.1 Data Collection and Preprocessing

- **Dataset:** Satellite imagery dataset was obtained from the INSAT-3D satellite series, operated by the Indian Space Research Organisation (ISRO). It provides high-resolution infrared imagery of cyclone events over the Indian Ocean region from 2013 to 2021. The dataset consists of thermal infrared channel images that capture cloud structures, cyclone formations, and intensity variations at regular intervals. Each image is time-stamped and associated with specific cyclone events, enabling detailed analysis for cyclone detection, classification, and forecasting tasks.
- **Data Retrieval:** The data utilized in this project was downloaded from Kaggle, namely the INSAT-3D Model dataset, from 2013 to 2023. The dataset contains high-resolution satellite images as well as meteorological data derived from the INSAT-3D satellite.
- **Preprocessing:** Preprocess the retrieved satellite imagery to enhance quality, remove noise, and standardize formats for further analysis. Techniques

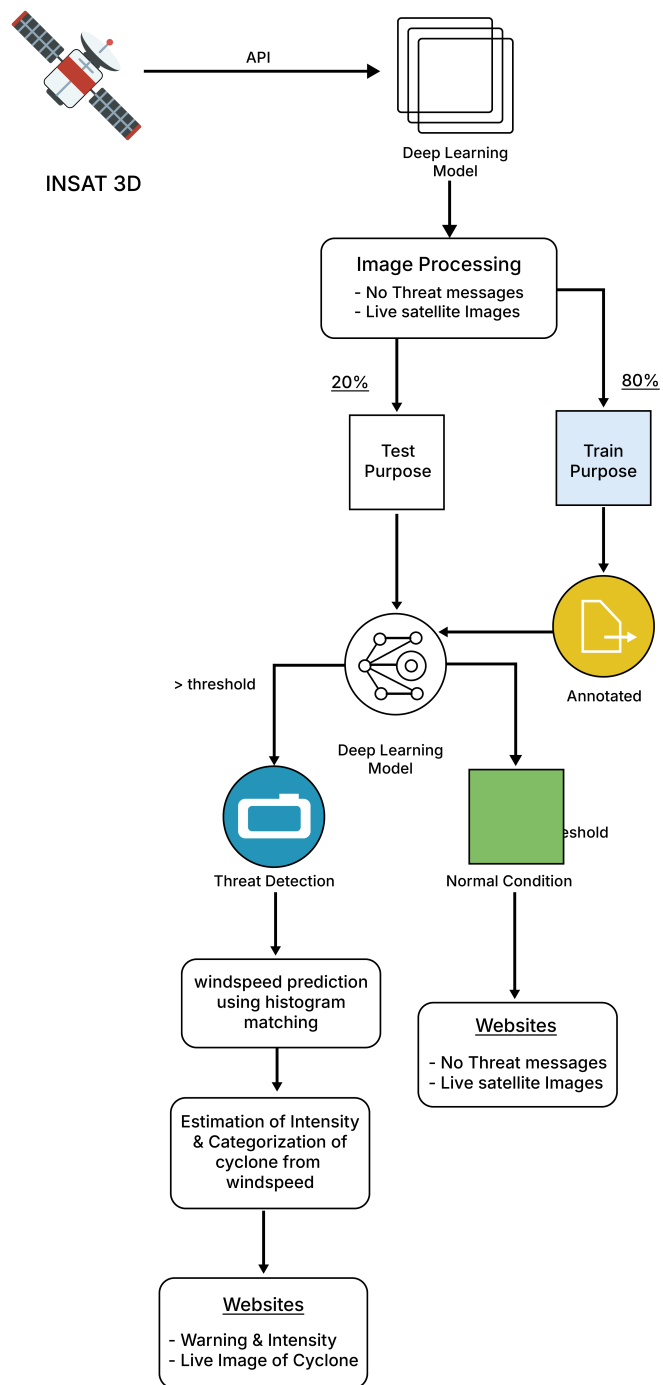


Figure 4.1: Proposed Workflow for Cyclone Detection

such as image normalization and enhancement may be applied.



## 4.2 Proposed Workflow

The proposed cyclone detection workflow begins with retrieving satellite images captured by the INSAT-3D satellite, which are accessed through kaggle dataset. These images undergo initial processing to filter out non-threatening conditions and normalization to reduce noise. The dataset is then split into two parts: 80% is used for training purposes and 20% for testing purposes. The training images are annotated to label cyclonic and normal conditions, which are then used to train a deep learning model. This model is designed to detect potential cyclone threats based on a defined threshold. If the model output exceeds the threshold, it indicates a threat detection; otherwise, the condition is marked as normal. Normal conditions are displayed on websites along with live satellite images and no-threat messages. In the case of threat detection, further processing involves predicting windspeed through histogram matching techniques. The estimated windspeed is used to assess the intensity and categorize the cyclone. Finally, websites are updated with warnings, intensity information, and live cyclone images, ensuring timely and accurate communication to users.

## 4.3 Design

### 4.3.1 Model Development

- **Convolutional Neural Network (CNN):** Design and train a CNN model using deep learning frameworks such as TensorFlow or PyTorch. The model architecture should be tailored for intensity estimation based on features extracted from satellite imagery.

- **Training:** Utilize the pre-processed satellite imagery dataset to train the CNN model, optimizing hyperparameters and regularization techniques to achieve optimal performance.
- **Testing:** Test the model using another distinct collection of image data that was not used for training or validation.

#### 4.3.2 Interface Development

The interface of the cyclone detection system is built using HTML, CSS, and JavaScript, while the backend is handled by Flask. HTML builds the web page, CSS formats it for a neat and responsive look, and JavaScript dynamically reloads the page by retrieving real-time cyclone detection information from Flask APIs.

# Chapter 5

## Results and Discussion

### 5.1 Experimental Setup

This study was conducted using Google Colab, a cloud environment with access to the high-end computer resources required to execute a deep learning program. These have been chosen as they are easier to use as well as needing the significant processing power. The environment presented by Google Colab had access to NVIDIA T4 GPUs providing 15.0 GiB each in memory, an on-cloud CPU, 12.67 GiB or more RAM, and up to 112.6 GB GiB of disk storage.

### 5.2 Dataset

The INSAT-3D Infrared RAW Cyclone Images dataset consists of satellite imagery collected from 2013 to 2021, specifically focusing on tropical cyclones over the Indian Ocean region. Captured by the INSAT-3D satellite operated by the Indian Space Research Organisation (ISRO), the dataset provides infrared channel images (IR1 and IR2 bands) that record thermal radiation from the Earth's surface and atmosphere. These images are particularly valuable for analyzing cloud patterns, cyclone

formation, and intensity variations over time. Each image is associated with a corresponding timestamp and cyclone event, making it suitable for time-series analysis, cyclone detection, and forecasting model development. The dataset offers a critical resource for researchers aiming to improve early warning systems, understand cyclone behavior, and develop machine learning models for environmental monitoring.

### 5.3 Performance Metrics

This study computed to quantify the model’s predictive accuracy. The Mean Absolute Error (MAE) measures the average deviation between predicted and actual intensity values, calculated as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where  $y_i$  is the ground-truth intensity and  $\hat{y}_i$  is the model’s prediction. The Root Mean Squared Error (RMSE), defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

emphasizes larger errors, providing insight into extreme prediction inaccuracies.

The dashboard serves as the primary interface for users, enabling seamless image uploads and visualization of predicted intensities overlaid on geographical maps. This tool enhances usability for non-technical stakeholders, such as emergency responders, by providing intuitive access to critical data.

The experimental validation underscores the system’s potential to improve disaster preparedness through accurate, real-time cyclone intensity

estimation, supported by robust metrics and scalable deployment architecture.

## 5.4 Results

The proposed deep learning framework for tropical cyclone detection, tracking, and intensity estimation demonstrated significant improvements over traditional methods. The key results are summarized as follows:

### 5.4.1 Image Processing

As shown in the Figure 5.1, during the model training phase, in this project, infrared satellite images have been used to train the cyclone detection model. Infrared images are crucial because they capture temperature variations in cloud patterns, which helps to detect cyclone formations even at night or under cloudy conditions.

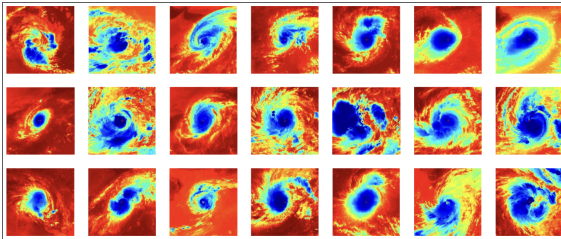


Figure 5.1: Infrared Images

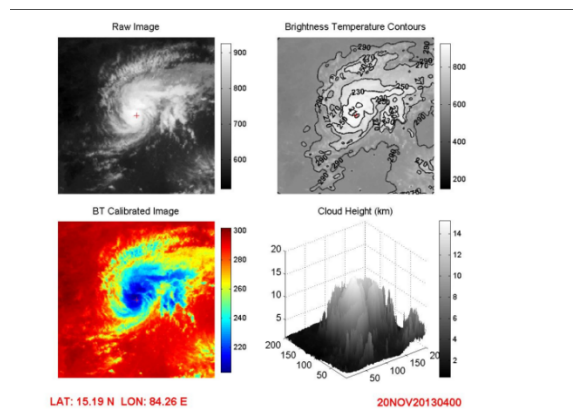


Figure 5.2: Cyclone Detection Image  
Description

### 5.4.2 Intensity Estimation of Cyclone using CNN Model

During the model training phase, Figure 5.3 shows the infrared satellite image (INSAT-3D) of a cyclone with the estimated intensity. The model predicts an intensity of 35.0 knots. The image size is  $250 \times 250$  pixels with 3 channels (RGB format after pre-processing). The red core indicates high cloud density and low temperature regions, suggesting a stronger cyclone center.

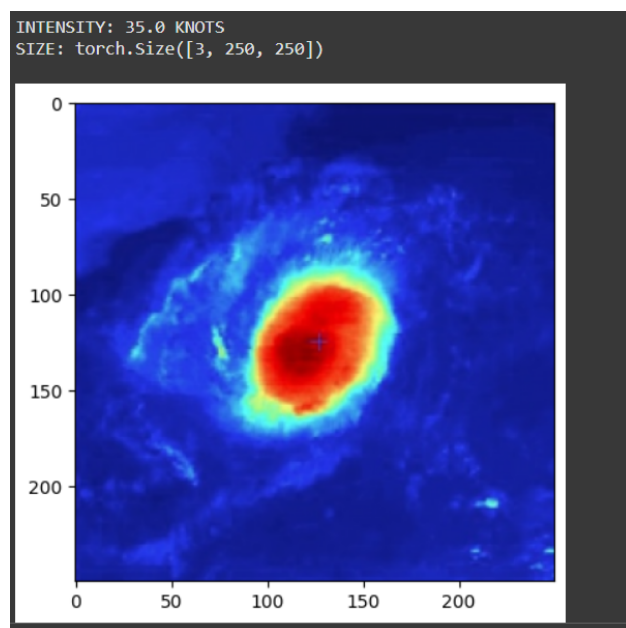


Figure 5.3: Intensity Estimation of Cyclone using CNN Model.

### 5.4.3 User-Friendly Interface

The Cyclone Detection System uses a Convolutional Neural Network (CNN) model integrated into a Flask Web Interface. The homepage (Figure 5.4) of the application provides users with two key options.

- The Live Weather Map (Figure 5.5): The "Live Weather Map" option displays a real-time map of the Indian Ocean, providing users with dynamic weather data. The map visualizes wind patterns and

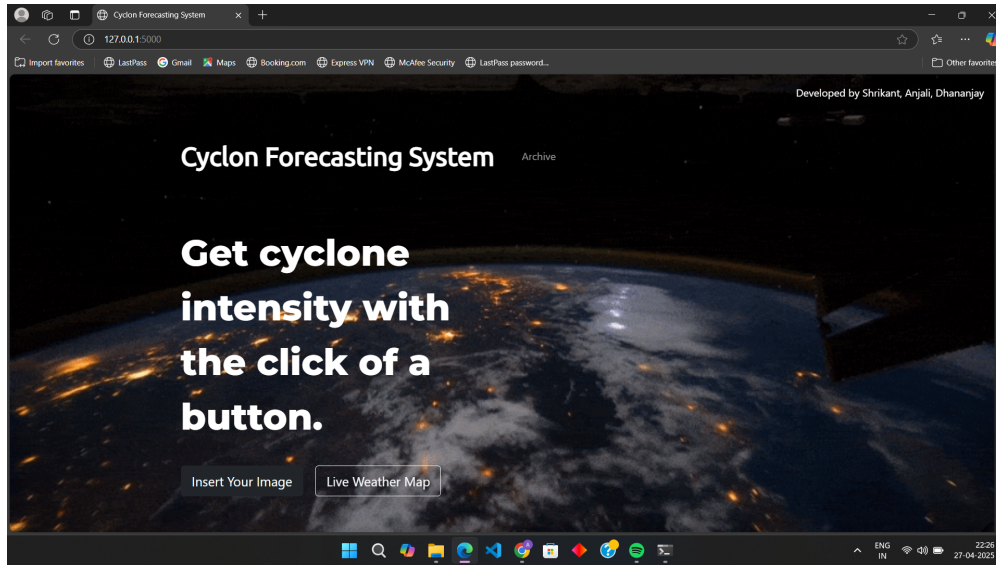


Figure 5.4: Home Page

tracks the development of cyclones. This feature allows users to monitor wind activity and identify whether a cyclone is forming based on real-time weather conditions.

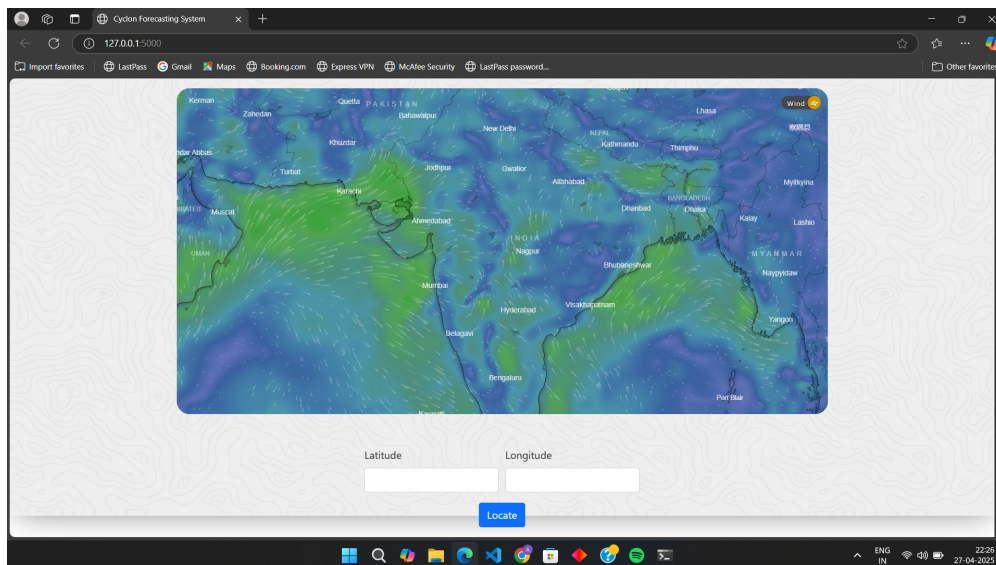


Figure 5.5: The Live Weather Map

- Insert Your Image (Figure 5.6): Users can upload satellite or weather images for analysis. The CNN model processes these images to detect and classify cyclone-related patterns, providing insights based on image recognition and deep learning techniques.

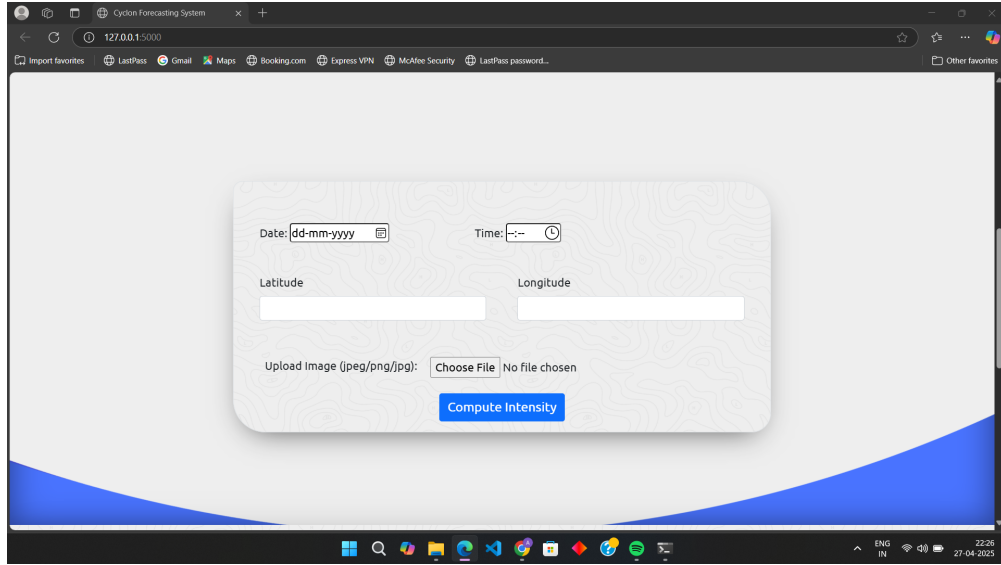


Figure 5.6: Insert Your Image

The evaluation metrics indicate the model’s performance in predicting cyclone intensity:

| Metric              | Value        |
|---------------------|--------------|
| Test RMSE           | 19.85 knots  |
| Test MAE            | 15.44 knots  |
| Test MSE            | 393.90 knots |
| R-squared ( $R^2$ ) | 0.19         |

Table 5.1: Model Performance Metrics for Cyclone Intensity Prediction

These results underscore the potential of the proposed framework to revolutionize tropical cyclone analysis. However, challenges such as handling data heterogeneity and reducing false positives in early cyclone detection remain areas for further refinement.



# Chapter 6

## Conclusion and Future Scope

The development and deployment of the cyclone forecasting system is a great achievement in disaster management, especially for the places that are usually affected by tropical storms. The use of advanced technology like satellite images, machine learning, and real-time data analysis makes it a great improvement in weather and cyclone prediction when it comes to the intensity of cyclones. This is so important since cyclones are a great danger to the coastal infrastructures, economy, environment, and people's lives.

In conclusion, the cyclone forecasting system is a critical implementation for enhancing preparedness for natural disasters. It is a forward-thinking and information-driven initiative that prioritizes the use of technology and teamwork among meteorologists and disaster management organizations. Incorporating this system into early warning systems helps to ensure that emergency evacuations and responses are timely and appropriate as well as to aid in long-term planning. It has the potential to save lives, reduce casualties, empower vulnerable populations, and significantly increase resilience to cyclones, among other benefits.

Research and development on the cyclone forecasting system is still ongoing.

ing, with future developments aimed at resolving existing issues and utilizing state-of-the-art technologies to improve the system's overall functionality and performance. Enhancing data fusion capabilities through the integration of data from various sources, including satellites, ground-based stations, and atmospheric simulations, is a crucial avenue. The objective is to better integrate this heterogeneous data to generate more accurate forecasts and improved situational awareness in real time.

The development of ensemble modeling, which combines several forecasting models, each with distinct advantages, to generate more reliable results, is another exciting avenue. Ensemble approaches can produce more robust and precise cyclone path and intensity forecasts by allocating dynamic weights according to each model's past performance.

Future research also suggests creating high-resolution numerical weather models especially for predicting cyclones. The goal of these models is to more accurately depict convective activity, storm surges, and localized weather patterns. Forecast accuracy will be further increased by utilizing better parameterization techniques and computational resources.

Furthermore, there is a lot of promise in combining machine learning and artificial intelligence (AI). Large volumes of complex data can be analyzed using tools like convolutional and recurrent neural networks, which improve the accuracy of pattern and trend identification. By enhancing lead times and automating the forecasting process, these intelligent systems can help disaster managers make decisions more quickly and intelligently.

# Appendix A

## Publication



Figure A.1: Book Chapter: Coastal Environments - Risk Assessment, Restoration and Sustainable Solution

The book chapter titled "Tropical Cyclone Detection and Intensity Estimation" is communicated to the book "Coastal Environments: Risk Assessment, Restoration and Sustainable Solutions" published by Springer Nature under the Sustainability Solutions book series.

# Appendix B

## Timeline

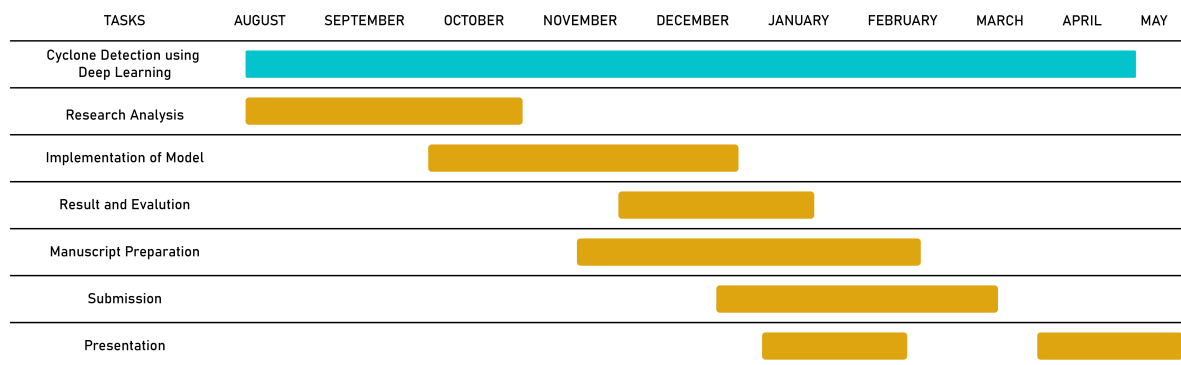


Figure B.1: Timeline Chart

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